



Smart Home IoT: Sensor Integration and Smart Control Based on ESP32

Final Project Report

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Program: Sistem Informasi Jaringan dan Aplikasi

Institution: SMK Negeri 7 Semarang

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Project Objectives & Core Vision

Core Purpose: To design and develop an integrated smart home system utilizing the ESP32 microcontroller.



Security

Enabling secure door access and active environmental monitoring.



Comfort

Automating physical responses to changing weather and climate.



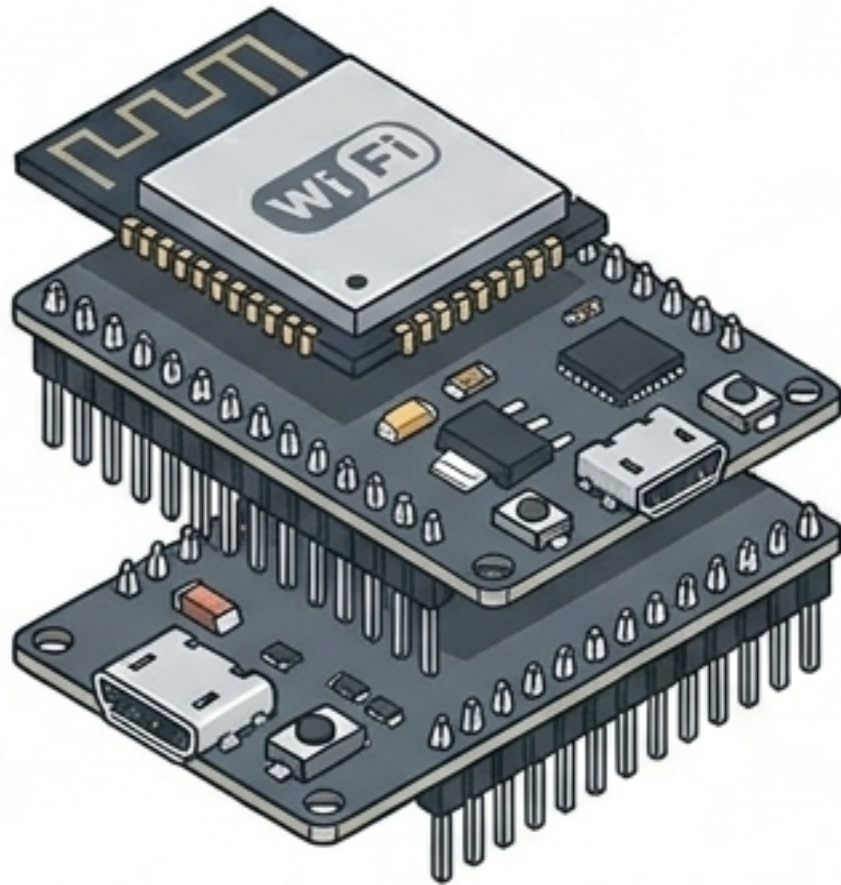
Efficiency

Creating smart resource management through remote integration.

Intended to improve home security, user comfort, and energy efficiency through real-time monitoring and remote control.

Hardware Ecosystem Architecture

The Brain (Central Processing)

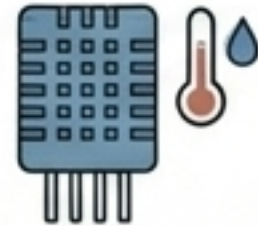


ESP32 Microcontroller

The Senses (Input Sensors)



PIR Sensor
(Motion detection)



DHT11 Sensor
(Temperature & Humidity)

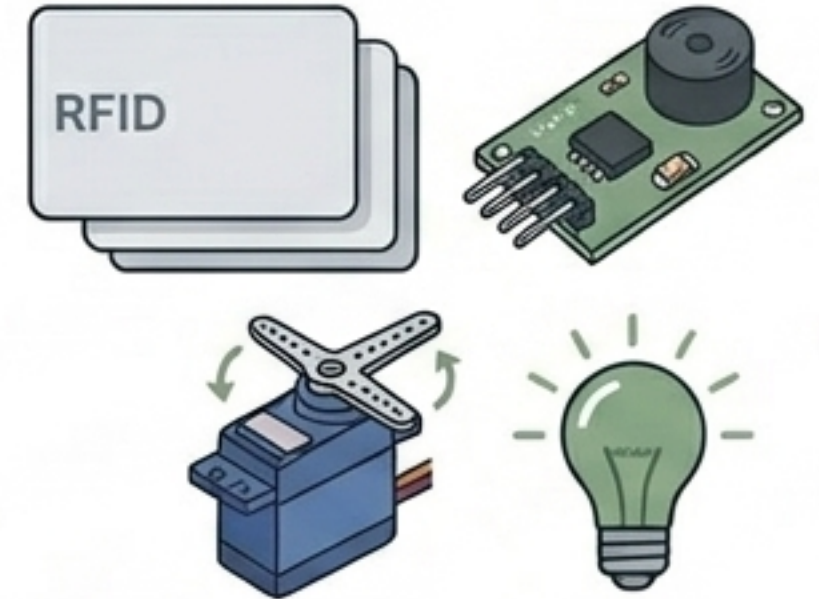


MQ-2 Gas Sensor
(Leakage detection)



Rain Sensor
(Weather monitoring)

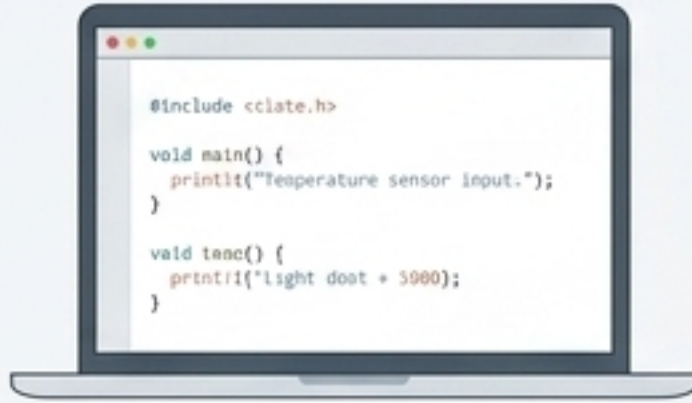
The Actions (Outputs & Access)



RFID Module & Card
(Secure access)

Servo Motors, Buzzer & LEDs
(Automated responses)

Software Ecosystem & Control Interface



Arduino IDE

- Used for writing, compiling, and uploading the core program to the ESP32.
- Processes sensor inputs and dictates automated hardware behaviors.

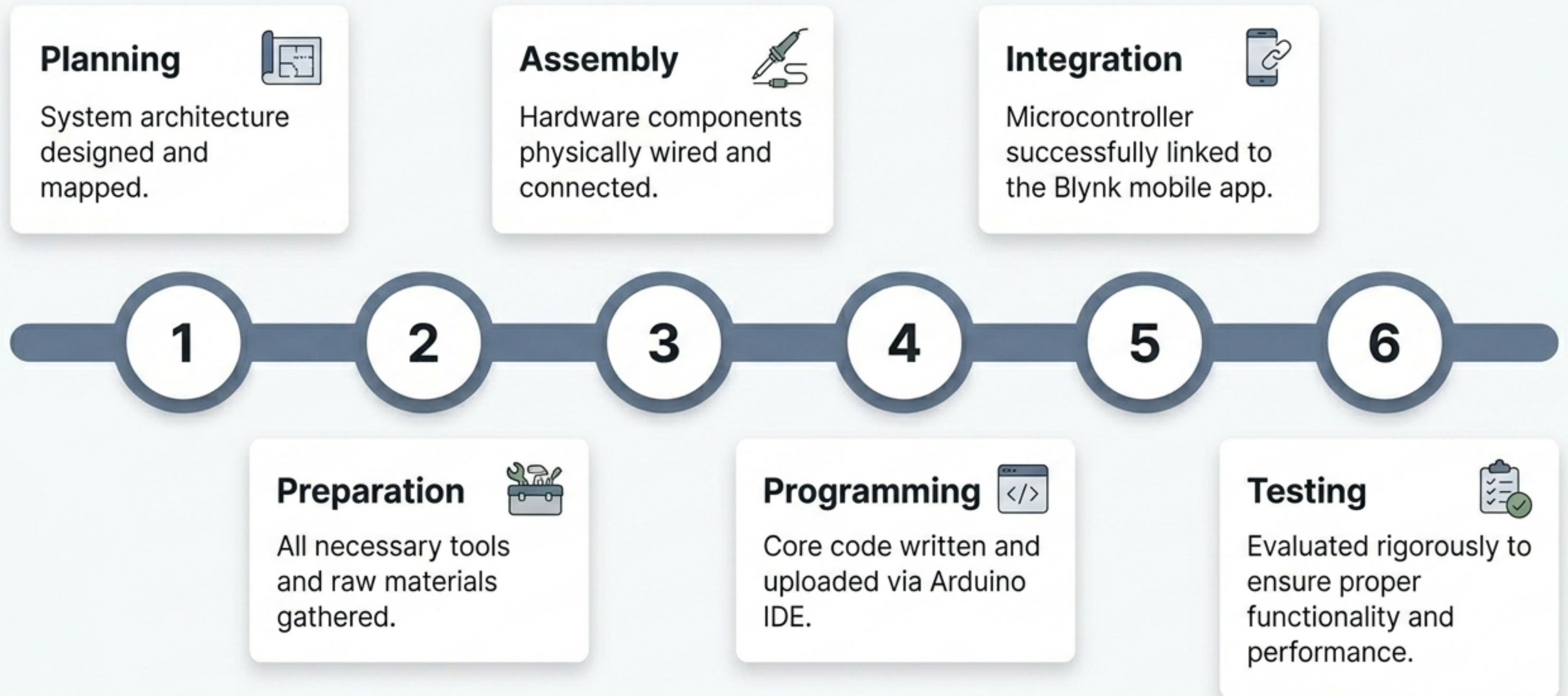


Blynk Application

- Smartphone-based remote control system.
- Transforms raw sensor data into an active, user-facing monitoring dashboard.

"Integrated with the ESP32 and connected to the Blynk application, allowing real-time remote monitoring and control."

6-Step Development Procedure



System Testing & Diagnostic Results



PIR Sensor

Detected motion effectively.



RFID Module

Controlled secure door access.



DHT11 Sensor

Monitored temperature and humidity accurately.



MQ-2 Sensor

Detected gas leakage reliably.



Rain Sensor

Automatically adjusted the clothesline via servo motors based on weather conditions.

System Health Summary: Demonstrated stable and reliable performance during testing.

Project Conclusion & Impact

“The project was successfully completed...
Resulting in an improved understanding of IoT
technology, electronics, and programming in a
smart home system.”



Functional Success

All physical components and software integrated seamlessly, operating exactly in accordance with the planned design.



Real-World Impact

Successfully delivered a proof-of-concept that improves home security, user comfort, and energy efficiency.



Educational Value

Provided profound practical experience bridging electronics, programming, and hardware integration.